MILITARY SPECIFICATION SHEET

NUT, PLAIN AND SELF-LOCKING, CLINCH, FIXED, METRIC (SELF CLINCHING, KNURLED COLLAR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the nut described herein shall consist of this specification and the latest issue of DOD-N-70333.

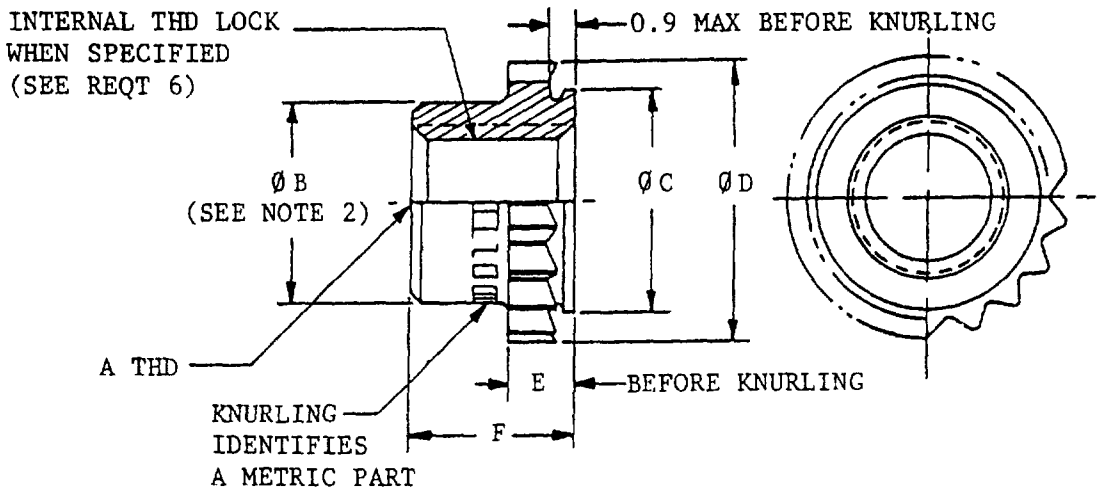


FIGURE 1. Dimensions and configuration.

NOTES:
1. Unless otherwise specified, dimensions are in millimeters and shall apply after plating and before the addition of solid film lubricant.
2. Dimension B is prior to deformation for self-locking nuts.

TABLE I. Part dimensions and dash numbers.

DASH NUMBER			A THD (SEE REQT 5)	ØB (SEE NOTE 2)	ØC	ØD	E	F	MASS Kg/1000 PCS APPROX REF		
ALLOY STEEL	CRES 303	CRES A286		±0.15	0 -0.15	±0.3	±0.15	±0.45	ALLOY STEEL	CRES 303	CRES A286
1A	1	1C	MJ2.5x0.45-4H6H	3.30	4.27	5.8	2.00	3.80	0.245	0.249	0.248
2A	2	2C	MJ3x0.5-4H6H	3.80	4.27	5.8	2.00	3.80	0.300	0.306	0.304
3A	3	3C	MJ3.5x0.6-4H6H	4.40	5.67	7.3	2.00	4.80	0.465	0.474	0.471
4A	4	4C	MJ4x0.7-4H6H	5.10	5.67	7.3	2.00	4.80	0.550	0.560	0.557
5A	5	5C	MJ5x0.8-4H6H	6.10	6.77	8.3	2.00	5.80	0.780	0.795	0.790
6A	6	6C	MJ6x1-4H5H	7.00	7.77	9.7	2.40	6.80	1.250	1.274	1.266

FSC 5310

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REQUIREMENTS:

1. Material: Alloy steel, grade 8740 (UNS G87400) conforming to AMS 6322 or grade 4140 (UNS G41400) conforming to AMS 6382. Corrosion-resistant steel, type 303 (UNS S30300), condition A or B conforming to ASTM A581. Corrosion-resistant steel, type A286 (UNS S66286) conforming to AMS 5731 or AMS 5734.

2. Protective finish and surface treatment: Alloy steel nuts (UNS G87400) or (UNS G41400) shall be cadmium plated; in addition, self-locking alloy steel nuts shall also have a solid film lubricant coating. Corrosion-resistant steel non-locking nuts (UNS S30300) or (UNS S66286) shall be cleaned, descaled and passivated. Corrosion-resistant steel (UNS S30300) self-locking nuts shall be cleaned, descaled and passivated followed by a solid film lubricant coating. Corrosion-resistant steel (UNS S66286) self-locking nuts shall be silver plated 5 micrometers (μm) thick plus a solid film lubricant coating.

3. Surface texture: Machined surfaces shall be 3.2 μm except knurling.

4. Hardness: Alloy steel (UNS G87400) shall be 44 to 50 HRC. Corrosion-resistant steel (UNS S30300) shall be 171 HB minimum. Corrosion-resistant steel (UNS S66286) shall be 34 HRC minimum.

5. Threads: Threads shall be in accordance with table I and general specification. Thread dimensions shall apply before application of solid film lubricant where applicable.

6. Military part number: Consists of the letter D, plus the basic number of the specification sheet and a dash number taken from table I. Add suffix letter L for self-locking nut. No letter is required for plain nut.

Example of Military part number.

D70333/1-	3C	L
Basic number of specification sheet	Dash Number	Self-locking (omit for plain nut)

Explanation of part number example:

D70333/1-3CL = Nut, Self-locking, Clinch, Fixed, Metric, MJ3.5x0.6-4H6H size thread, pitch and tolerance class, CRES A286, silver plated plus a solid film lubricant coating.

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7. Resistance to push-out and torque-out: Push-out and torque-out values shall be as specified in table II herein when nut is properly installed in aluminum, type 6061-T6 (UNS A96061) in accordance with QQ-A-250/11, Temper T6.

TABLE II. Push-out and torque-out values.

INT THD SIZE	PUSH-OUT MIN N	TORQUE-OUT MIN N·m
MJ2.5x0.45	500	2.6
MJ3x0.5	500	2.6
MJ3.5x0.6	520	4.5
MJ4x0.7	520	4.5
MJ5x0.8	860	6.7
MJ6x1	1200	9.6

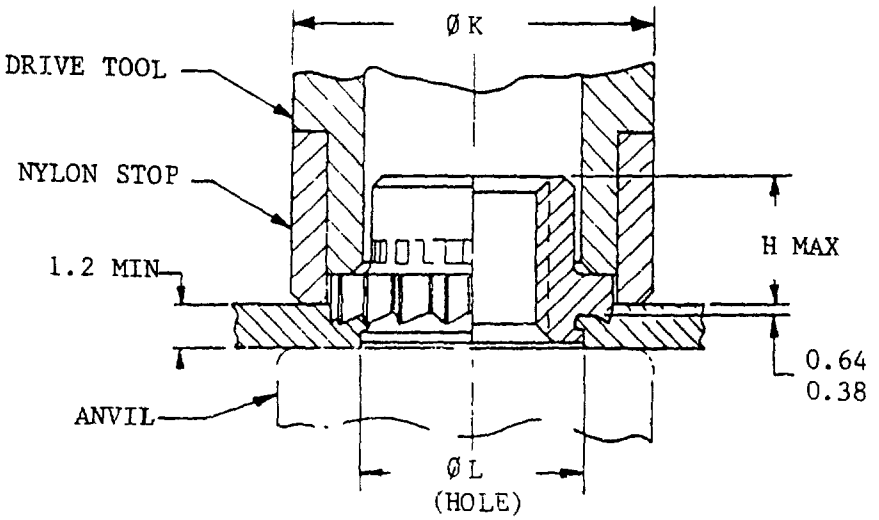


FIGURE 2. Hole preparation and installation.

TABLE III. Installation data in parent material.

DASH NUMBER			H *	ØK	ØL
ALLOY STEEL	CRES 303	CRES A286	MAX	REF	+0.1 0
1A	1	1C	2.9	14.2	4.3
2A	2	2C	2.9	14.2	4.3
3A	3	3C	3.9	15.8	5.7
4A	4	4C	3.9	15.8	5.7
5A	5	5C	4.9	15.8	6.8
6A	6	6C	5.9	19.0	7.8

* H dimension is for design clearance information only. Do not use as installation data. The manufacturer drive tool shall control the nut installation depth.

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Installation procedures.

a. Drill or punch hole (Corrosion-resistant steel shall be drilled).

a.1 Do not chamfer, break or deburr the hole on the side of the material where the nut will be installed .
a.2 In punched holes the nut shall be installed on the side where the punch exits.

b. Locate pilot squarely in hole.

c. Using a drive tool and anvil (as shown in Figure 2) apply sufficient force to cause a 0.38-0.64 mm penetration into the structure for a full 360°. Proper penetration is accomplished when nylon stop bottoms on surface of parent material. Do not overdrive the nut.

d. Install screw or bolt from pilot side of nut.

Custodians:
Army - AR
Navy - AS
Air Force - 99

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Air Force - 82
NSA - NS
DLA - IS

User activities:
Navy - MC

Preparing activities:
Army - AR

(Project 5310-1326)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
DOD-N-70333/1	

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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